

3

Problem 1

A particle of charge q and mass m is subject to a constant and uniform external electromagnetic field $F_{\mu\nu}$. Let $U^\mu(0)$ be the particle's initial 4-velocity at proper time $\tau = 0$, with $U^\mu U_\mu = -c^2$.

(a) Show that

$$U^\mu(\tau) = \left(e^{\tau \frac{q}{mc} F} \right)^\mu{}_\nu U^\nu(0).$$

HINT: Expand the exponential of a matrix in Taylor series to find the action of $\frac{d}{d\tau}$ on the exponential.

We can start from the answer and go back.

to start, let's call

$$k = \frac{q}{mc}$$

$$(e^{k\tau F})^\mu{}_\nu = \left(I + \tau k F + \frac{(\tau k F)^2}{2} + \frac{(\tau k F)^3}{3!} \dots \right)^\mu{}_\nu$$

We can see that

$$\frac{d}{d\tau} e^{k\tau F} = kF + \frac{\tau k^2 F^2}{1} + \frac{\tau^2 (kF)^3}{2!} + \dots$$

We can write this as a function of sinh and cosh

$$\frac{du^\mu}{d\tau} = kF \left(1 + \tau kF + \frac{\tau^2 (kF)^2}{2!} + \dots \right)$$

This becomes

$$a = kF(\sinh(\tau kF) + \cosh(\tau kF))$$

$$F = ma \stackrel{?}{=} qu^\mu F^{\mu\nu}$$

We see that we are proportional to u , since this is just kF times the expanded version of u^μ , so we have

$$F = kFmu^\mu$$

$$F = \frac{q}{c} F u^\mu$$

We have shown what we wanted to show.

Old work

This gives us the velocity over time as

We are trying to show that the acceleration makes the function of time

$$F = ma$$

$$u(\tau)^\mu = u^\mu(0) + m \int_0^\tau a^\mu(t) dt$$

Where

$$u^\mu(\tau) = \int_0^\tau kF(\sinh(\tau'kF) + \cosh(\tau'kF))d\tau'$$

$$u^\mu(\tau) = kF(\cosh(\tau kF) + \sinh(\tau kF)) \Big|_0^\tau$$

$$u^\mu(\tau) = kF(e^{k\tau F}) - kF$$

(b) Find $U^\mu(\tau)$ for the case where

$$\vec{E} = (E, 0, 0) \quad \vec{B} = (B, 0, 0)$$

which can always be arranged by a Lorentz transformation except if $E = B$ and $\vec{E} \perp \vec{B}$. HINT: First show that the matrix in the exponent is block diagonal. Then expand in Taylor and combine the infinite series in sines, cosines, sinhs, and coshs.

Let

$$\bar{\sigma} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

These are found from writing the fields in some way.

$$e^{\frac{\sigma}{mc}F} = \begin{pmatrix} \cosh(\epsilon\tau)1_{2 \times 2} + \sinh(\tau\beta)\sigma_{2 \times 2} & 0 \\ 0 & \cos(\beta\tau)1_{2 \times 2} + \sin(\beta\tau)\sigma \end{pmatrix}$$

(c) Set $B = 0$ in (b), let

$$U^\mu(0) = \frac{1}{\sqrt{1 - v_0^2/c^2}} (c, 0, v_0, 0).$$

Find $\vec{x}(t)$ by integrating your equations from (b). Show in particular that, if $v_0 = 0$,

$$x^1(t) = \sqrt{(ct)^2 + \frac{m^2 c^4}{q^2 E^2}}, \quad x^2(t) = x^3(t) = 0.$$

What is the maximum speed of the particle when $t \rightarrow \infty$?

Let

$$\epsilon = \frac{Eq}{mc}$$

We can set a bunch of things equal to zero, and say

$$u^\mu(\tau) = \begin{pmatrix} \cosh(\epsilon\tau) & \sinh(\epsilon\tau) & 0 & 0 \\ \sinh(\epsilon\tau) & \cosh(\epsilon\tau) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^\mu u^\nu(0)_\nu$$

We have the identity and integrate with respect to τ .

$$x^\mu(\tau) = \frac{1}{\epsilon} \begin{pmatrix} \sinh(\theta\tau) & \cosh(\epsilon\tau) - 1 & 0 & 0 \\ \cosh(\epsilon\tau) - 1 & \sinh(\theta\tau) & 0 & 0 \\ 0 & 0 & \tau & 0 \\ 0 & 0 & 0 & \tau \end{pmatrix}^\mu u^\nu(0)_\nu \begin{bmatrix} c \\ 0 \\ v_0 \\ 0 \end{bmatrix} \frac{1}{\sqrt{1 - \frac{v_0^2}{c^2}}}$$

We have

$$X^0(\tau) = ct = \frac{c}{\sqrt{1 - \frac{v_0^2}{c^2}}} \frac{1}{\epsilon} \sinh(\epsilon\tau)$$

$$x'(\tau) = \frac{1}{\epsilon \sqrt{1 - \frac{v_0^2}{c^2}}} (\cosh(\epsilon\tau) - 1)$$

Lets call

$$G_{\text{arbage}} = \frac{1}{\epsilon \sqrt{1 - \frac{v_0^2}{c^2}}}$$

We can divide by garbage, and have

$$\frac{x'(\tau) + G}{G} = \cosh(\theta\tau)$$

$$\frac{x^0(\tau)}{G} = \sinh(\epsilon\tau)$$

We have

$$\cosh^2 - \sinh^2 = 1$$

$$\left(\frac{x+g}{G}\right)^2 - \frac{c^2 t^2}{G^2} = 1$$

What is the maximum speed of the particle when $t \rightarrow \infty$?

Problem 2

Use Green's 1st theorem to show that the solution to $\nabla^2 \phi = -4\pi\rho$ is unique with fixed Dirichlet or Neumann boundary conditions. Do this by assuming two different solutions ϕ_1 and ϕ_2 , and showing $\phi_1 = \phi_2$ by solving $\nabla^2(\phi_1 - \phi_2) = 0$.

HINT: Set $\varphi = \psi = \phi_1 - \phi_2$ in Green's first theorem

$$\int (\varphi \nabla^2 \psi + \vec{\nabla} \varphi \cdot \vec{\nabla} \psi) dV = \oint \varphi \frac{\partial \psi}{\partial n} da$$

If we have two solutions that satisfy laplace's equation,

$$\nabla^2 \phi_1 = -4\pi\rho,$$

$$\nabla^2 \phi_2 = -4\pi\rho$$

then

$$\nabla^2(\phi_1 - \phi_2) = 0$$

let $\psi = \phi = \phi_1 - \phi_2$

in Green's theorem,

$$\int (\phi \nabla^2 \phi + \vec{\nabla} \phi \cdot \vec{\nabla} \phi) dV = \oint \phi \frac{d\phi}{dn} da$$

If we are assuming $\phi = 0$, then we need to show that

$$\oint \phi \frac{d\phi}{dn} da = 0$$

$\phi_1 = \phi_2$ on the surface is set by the boundary condition.

$\frac{d\phi_1}{dn} = \frac{d\phi_2}{dn}$ on the surface is also set by the boundary condition., so

$$\frac{d\phi}{dn} = 0 \text{ or } \phi = 0$$

Problem 3

Use Gauss' law to find the electric field inside and outside a sphere of radius R and charge density

$$\rho = \alpha r^n \quad (\text{with } n > -3).$$

Write the answer in terms of the total charge Q .

$$\oint \oint_s \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

The total charge enclosed by a hypothetical sphere of radius r is

$$Q_{enc} = \int_{r'=0}^r \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\frac{\pi}{2}} \alpha r^n (r dr d\theta d\phi)$$

because of the spherical symmetry of the problem, this is just a thin shell of surface area $4\pi r^2$ with charge $r\pi r^2 \rho(r)$. the total charge is then

$$\int_{r=0}^r 4\pi \alpha r'^{n+2} dr'$$

The condition $n > -3$ lets us integrate this with a reasonable form.

This integrates to

$$Q_{enc} = 4\pi \alpha \frac{r^{n+3}}{n+3}$$

If we enclose the entire sphere, the limit of Q_{enc} is

$$Q_{enc} = 4\pi \alpha \frac{R^{n+3}}{n+3}$$

Knowing Q_{enc} , we know have $\vec{E}$, since $\vec{E}$ is everywhere perpendicular to the surface of the sphere (we know by symmetry).

$$\vec{E} \cdot 4\pi r^2 = \frac{Q_{enc}}{\epsilon}$$

$$\vec{E} = \frac{Q_{enc}}{4\pi r^2 \epsilon_0}$$

This gives us the two cases:

$$\vec{E} = \frac{4\pi \alpha}{n+3} \begin{cases} R^{n+3} & r \geq R \\ r^{n+3} & r < R \end{cases}$$

Problem 4

Calculate the capacitance of

- (a) Two large plates of area A and separation $d \ll \sqrt{A}$.
- (b) Two concentric spheres of radii a and b ($b > a$).
- (c) Two concentric cylinders of length L and radii a, b ($b > a$).

Superposition of each plate's field, Gauss's law.

We place a charge $+Q$ on one plate, and $-Q$ on the other plate, so that there is no net electric field outside the capacitor. For each scenario, these are our boundary conditions.

a)

the distance is much less than the length of the sheet, so we can approximate the capacitor as infinitely wide.

We can enclose a cube through each of the two plates of the capacitor, one for each side, and add together the solutions as a superposition of the charges.

$$\Phi_E = \oint E \cdot dA = \frac{Q_{enc}}{\epsilon_0}$$

The electric field is perpendicular to the top / bottom ends of the cube everywhere, and parallel to the sides. If s is the side length of the square face of the rectangle, the flux is just

$$Q_{enc} = \sigma s^2$$

$$E \cdot 2s^2 = \frac{Q_{enc}}{\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}$$

Each of the two plates has this field, so we have

$$E_+ = \frac{\sigma}{2\epsilon_0}$$

$$E_- = -\frac{\sigma}{2\epsilon_0}$$

The positive field is pulling negative charges towards itself, and the negative field is pushing away from itself. The fields must therefore have a higher amplitude between them (it can't be zero), so we have

$$E = \frac{\sigma}{\epsilon_0}$$

Capacitance is the charge per volt, and the voltage is the potential difference between the plates, the work done to bring a charge Q from one to the other

$$V = \int_0^d \frac{\sigma}{\epsilon_0} dz = \frac{\sigma d}{\epsilon_0}$$

Capacitance is

$$C = \frac{Q}{V} = \frac{\epsilon_0 Q}{\sigma d}$$

we said that $Q = \sigma s^2$, so this is

$$\frac{\epsilon_0 s^2}{d}$$

$$= \frac{\epsilon_0 A}{d}$$

b

(b) Two concentric spheres of radii a and b ($b > a$).

If we draw a Gaussian surface sphere around the inner one, we would have charge Q . If we have charge $-Q$ on the outer one, then we would have no net charge enclosed when encapsulating both. The field thus only exists from one plate with charge Q , not from the other.

With the spherical symmetry, we always radiate radially out. Thus, we have

$$E \cdot 4\pi a^2 = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi a^2 \epsilon_0}$$

This gives us

$$V = \int_a^b \frac{Q}{4\pi \epsilon_0} \frac{1}{a^2} dr$$

$$= \frac{Q(b-a)}{4\pi \epsilon_0} \frac{1}{a^2}$$

so the capacitance is

$$\frac{4\pi \epsilon_0 a^2}{(b-a)}$$

(c) Two concentric cylinders of length L and radii a, b ($b > a$).

We can have a closed outer gaussian shape that doesn't pierce the surface of the outer cylinder and still doesn't enclose the inner one. Therefore we should (probably) have a superposition of both shapes. I'll put Q on the inner cylinder, and $-Q$ on the outer one.

Lets assume that L is large compared to a, b . Lets make a gaussian cylinder around the inner cylinder.

$$E \cdot 2\pi a L = \frac{Q}{\epsilon_0}$$

This is also constant, so we have

$$V = \int_a^b \text{stuff } dr$$
$$V = \frac{Q}{2\pi L \epsilon_0} \left(\frac{1}{a} \right) (a - b)$$

and capacitance

$$C = \frac{2\pi L a \epsilon_0}{a - b}$$